

Sub-THz Atmospheric Sensing with Pilot and Payload Reuse: Recall Gains and Calibration Limits

Guillem Moreno Garcia
Universitat Politècnica de Catalunya
Email: guillem.moreno.garcia@estudiantat.upc.edu

Abstract—We evaluate atmospheric absorption sensing in a 60–400 GHz satellite downlink using externally parameterized HITRAN spectroscopy, refracted paths, microwave emission, particle distributions and explicit radio resources. Observations remain simulated. The original favorable broad-reference recall assumes zero residual calibration error after correction, an accuracy not demonstrated experimentally. Processing existing constant modulus payloads with second/fourth moment signal and noise separation addresses the information loss from using only 30 pilots per 10,000 symbols. At 1 microgram per cubic metre, 10 s and zero persistent calibration residual, acetonitrile recall rises from 0.460% to 99.955% with unchanged transmissions and decision threshold. An independent simulated 10 s reference plus 10 s sensing gives 93.800% enhancement recall. Formaldehyde reaches 95.275% at 100 s; methanol and joint particulate sensing remain weak. A sequential receiver extension adds raw moving coded OFDM, charged retuning and nonzero calibration controls. A revised five-gas band design gives 97.67% simulated acetonitrile recall at 1 microgram per cubic metre, 20 s total and an assumed, unmeasured 0.0001 dB residual. Full time/calibration tables retain weak gas and PM results and demanding RF requirements. Calibration, weather and modulation stress expose important limits. The ideal multiband gains do not transfer to the existing contiguous E-band radio. Public water and aerosol measurements support limited physical checks but do not validate atmospheric VOC retrieval. We report conditional sensing improvements and the measurements required to establish field performance.

Index Terms—Atmospheric sensing, volatile organic compounds, particulate matter, HITRAN, calibration, integrated sensing and communication.

I. QUESTION AND EVIDENCE

Differential absorption has been studied for water vapour in sub-terahertz satellite contexts [1]; multi-medium propagation models also motivate joint gas and particle analysis [2]. We ask whether a specified satellite channel can distinguish weak VOC and PM signatures while preserving its communication resources.

Implementing a model, evaluating an estimator, and demonstrating an environmental monitor are different milestones. An evaluation may legitimately produce a negative feasibility result. A successful numerical optimizer or a small local error bound does not establish a calibrated measurement.

This revision supplies an expanded atmosphere and spectroscopy implementation, explicit detection probabilities, nonlinear-response checks, OFDM waveform and moving-pass controls, a PM information diagnosis, and bounded comparisons with public measured evidence. All sensing calculations

TABLE I
EVIDENCE TYPES AND SUPPORTED INTERPRETATIONS.

Evidence	Interpretation
HITRAN catalogue	External spectroscopy; not measured atmospheric CSI.
Beijing surface data	Measured scenarios; not vertical pollutant profiles.
IGRA soundings	Measured weather; upper continuation is modeled.
Simulated pilots	Conditional estimator and detector controls.
Published 380 GHz results	Independent aggregate water-background check.
Public calcite THz-TDS	Measured cell response/drift; no PM mass labels.

remain in 60–400 GHz. The previous September 8 manuscript is preserved as a dated historical artifact.

II. ATMOSPHERE, SPECTROSCOPY AND PARTICLES

A. Weather and concentration profiles

The reference atmosphere implements ITU-R P.835-7 Annex 1 through 100 km, including height conventions and atmospheric interfaces [3]. NOAA IGRA data supply Beijing pressure, temperature, height and dew-point depression [4]. Parsing retains 103,322 levels from 521 soundings. A fixed rule selects the first complete profile reaching at least 20 km in January, April, July and September 2026. Measured tops range from 28.8 to 37.0 km above the station. Temperature and logarithmic pressure/vapour are interpolated; continuation above the measured top is explicitly modeled.

Weather soundings do not measure VOC or PM abundance profiles. Gas and dry-particle concentrations remain proportional to $\exp(-z/H)$ with declared scale heights of 1,500 and 1,000 m. The four weather profiles are examples, not a population distribution, and are not paired with the historical PM record or radio observations. Horizontal homogeneity and nondispersive ray refraction remain assumptions.

B. Spectroscopy and refracted integration

The acquired inventory contains 34 isotope tables, 304,416 catalogue transitions and 31,474 transitions in 60–400 GHz. Formaldehyde, methanol and acetonitrile are the VOC targets; CO, O₃, SO₂ and NO₂ are interfering gases. Methane is retained as an acquisition control. Out-of-band line centres contribute tails only. Three additional isotope/window requests

were unavailable and remain failed requests, not zero absorption.

For molecule s , the local absorption cross-section is

$$\sigma_s(\nu, z) = \sum_{\ell \in s} S_\ell(T(z)) V_\ell[\nu - \nu_\ell - \delta_\ell p(z); T(z), p(z)], \quad (1)$$

with temperature-dependent line strengths, partition sums, pressure shifts and air/self broadening [5], [6]. Voigt profiles combine pressure and Doppler effects. Each isotope uses its mass and partition sum; terrestrial abundance is not multiplied twice. Natural-mixture molar masses define reported mass/ppm conversions.

The attenuation is

$$A_s(\nu) = \frac{10}{\ln 10} \int_{\text{ray}} n_s(z) \sigma_s(\nu, z) ds. \quad (2)$$

Bouguer's invariant $b = n(r)r \sin \zeta$ defines a spherical refracted path connecting the receiver and satellite. Quadrature respects atmospheric interfaces, replacing a secant multiplication of a completed vertical column.

P.676-13 Annex 1 supplies microwave oxygen/water absorption, line-mixing and continuum terms [7]. It replaces HITRAN $\text{H}_2\text{O}/\text{O}_2$ in the current background, avoiding double counting. Twelve additional rare-isotope comparisons against HAPI pass with maximum peak-normalized discrepancy about 7.70×10^{-6} . HAPI shares the line data: this checks implementation, not independent physical truth. Missing broadening and parameter-uncertainty distributions limit unconditional claims. Far-wing truncation remains a sensitivity control.

C. Particles, emission and numerical checks

Full homogeneous-sphere Mie extinction is integrated over truncated lognormal number distributions and normalized by dry mass [8]. Fine/coarse modes use aerodynamic cuts and Stokes/Cunningham conversion; PM_{10} is their sum. A supplied humidity-growth interface has an explicit effective-medium assumption. Target aerosol optical constants, composition, shape and humidity response remain uncalibrated.

Rayleigh-versus-Mie relative RMS differences are approximately 0.00086% for fine particles and 0.0201% for coarse particles. Agreement in this size/frequency regime does not add independent measurement information.

With $J_\nu(T) = (h\nu/k)/[\exp(h\nu/kT) - 1]$, atmospheric thermal transfer gives

$$T_{\text{sky}} = \sum_i J_\nu(T_i)(1 - e^{-\tau_i})e^{-\sum_{j < i} \tau_j} + J_\nu(T_{\text{space}})e^{-\sum_i \tau_i}. \quad (3)$$

Receiver noise is $kB[T_{\text{sky}} + T_0(F - 1)]$; noise figure is not counted twice.

The numerical acceptance criterion is relative vector RMS difference below 0.1% between two quadrature orders. The initial thermal comparison failed at 0.11596%. Refining intervals to 100 m below 20 km and 500 m above reduced it to 0.00319%, without changing the tolerance or other physics arrays. Gas, PM and background differences are 0.00381%, 0.0225% and 0.00200%. An independent recursive transfer

calculation reproduces the refined brightness. This establishes numerical stability on the tested grid, not atmospheric accuracy. The correction changes broad-reference rate by only about -0.000270% .

The four measured-weather spectra use order two. Only the September profile has a separate order comparison, at 0.05304% relative RMS; four independently converged weather runs are not claimed.

III. RADIO AND INFERENCE MODEL

A. Resources and pilot observations

The reference link uses a 550 km satellite, 45° elevation, 23 dBm total power, 0.50/0.30 m transmit/receive apertures, 6 dB noise figure and 5 dB implementation loss. The broad case has 256 separated frequencies in 60–400 GHz and is an ideal simultaneous multiband experiment.

The explicit E-band case has 256 contiguous 1 MHz tones centred at 73.5 GHz, one RF chain, a 1/16 cyclic prefix and 30 pilots per 10,000-symbol frame. Its 73.372–73.628 GHz interval lies within the 71–76 GHz space-to-Earth reference discussed by ITU [9]. This is not a coordination or licensing determination. WR12 laboratory VNA extenders cover this interval but are not flight-modem validation [10].

After reference normalization and assumed phase tracking, the complex pilot mean is

$$\bar{z}_i = 10^{-[D\theta + N\beta]_i/20} + \eta_i, \quad \eta_i \sim \mathcal{CN}(0, (N_p \gamma_i)^{-1}), \quad (4)$$

where D holds three VOC and two PM signatures, and N holds background/calibration and interfering-gas terms. Magnitude-derived attenuation is $-20 \log_{10} |\bar{z}_i|$. Its local thermal variance is $2(10/\ln 10)^2/(N_p \gamma_i)$. Separate scenarios add persistent correlated residuals. Covariance and SNR are conditional on the reference channel. The retained 5 dB single-pilot SNR mask is not a universal sensing threshold.

B. Estimation, bias and detection

For $y = D\theta + N\beta + n$, set $LL^T = \Sigma$, $\tilde{D} = L^{-1}D$, and let Q span $L^{-1}N$. With $P = I - QQ^T$, the efficient local estimator is

$$\hat{\theta} = Hy, \quad H = (\tilde{D}^T P \tilde{D})^{-1} \tilde{D}^T P L^{-1}. \quad (5)$$

Its covariance attains the nuisance-adjusted Gaussian bound when the target is identifiable [12]. SVD implementation and checks of retained column size prevent a removed signature from being normalized into false information.

The nonlinear diagnostic fits the exact complex mean, with Gaussian real/imaginary samples and concentrations bounded to 0–1,000 $\mu\text{g m}^{-3}$. It retains optimizer status, active bounds and profile likelihoods. Unknown phase tracking is not silently solved. Signed estimates, rather than clipped values, determine reported error and detection.

For persistent $\|b\|_\infty \leq \epsilon$, the local worst-case concentration bias is $B_j = \epsilon \|h_j\|_1$. More pilots do not remove this term. Earlier Gaussian attainability work covers 48 configurations with 10,000 draws each and demonstrates strong sensitivity to persistent model mismatch; it remains archived.

TABLE II
BROAD 10 S REFERENCE. LOD ASSUMES 95% POWER AND 1%
FAMILY-WISE FALSE POSITIVES. RECALL AND RMSE USE $1 \mu\text{g m}^{-3}$
INJECTIONS. PPM USES 288.15 K AND 101325 PA.

VOC	LOD ($\mu\text{g m}^{-3}$)	ppm	Recall (%)	RMSE ($\mu\text{g m}^{-3}$)
Formaldehyde	56.21	0.04426	0.30	12.44
Methanol	131.24	0.09685	0.20	28.71
Acetonitrile	12.94	0.00745	0.20	2.83

Six reported targets include three VOCs, fine PM, coarse PM and PM_{10} . Bonferroni allocation uses family-wise false-positive budget $\alpha = 0.01$ and required power 0.95. With standard error s_j ,

$$c_{\text{crit},j} = z_{1-\alpha/6} s_j + B_j, \quad (6)$$

$$c_{\text{LOD},j} = (z_{1-\alpha/6} + z_{0.95}) s_j + 2B_j. \quad (7)$$

The factor two covers opposite worst-case biases under null and alternative. Bonferroni permits dependent PM-derived targets. This rule is stricter than the previous 1% per target and is unrelated to the numerical integration tolerance.

The grid attempts two bands, three nuisance policies, three durations (0.1, 10 and 1,800 s), and two residual levels: 36 configurations. Each evaluated case has 5,000 present and 5,000 absent controls. VOC injections are $1 \mu\text{g m}^{-3}$; fine/coarse PM are 49/41 $\mu\text{g m}^{-3}$ from one recorded Beijing scenario [13]. These are response controls, not an environmental population. Precision assumes artificial 50% prevalence. The retained tables have 114 detection, 228 error and 342 decision-limit rows; failed configurations are preserved.

IV. CONDITIONAL SENSING RESULTS

A. VOCs and nuisance effects

Table II uses the 10 s ideal broad reference, reference nuisance calibration, zero persistent residual and zero bias allowance. Recall at $1 \mu\text{g m}^{-3}$ is very low. At the larger predicted detection limits, new exact-response checks achieve 95.16%, 95.40% and 94.92% recall, within the declared one-percentage-point tolerance of 95%. These checks support three modeled operating points, not every uncertainty scenario or field performance.

Weather and smooth-instrument nuisance terms remove additional information. Seventeen of 18 E-band configurations fail the numerical audit. The remaining 1,800 s case has formal limits far beyond meaningful atmospheric interpretation and exceeds a single pass. Its marginal numerical rank does not establish useful sensing. The result rejects this evaluated design, not every possible E-band waveform.

B. PM information and auxiliary requirements

Formal local PM detection limits are about 4.85×10^8 , 5.96×10^8 and $1.10 \times 10^8 \mu\text{g m}^{-3}$ for fine, coarse and total PM. These are failure diagnostics, not validated concentration ranges. All 20 nonlinear fits report success, yet fine PM reaches a bound in 100% and coarse PM in 95%. The fine-PM likelihood is nearly flat across the fitted domain.

TABLE III
OFDM CONTROLS WITH THE SAME SEED ACROSS IMPAIRMENT CASES.
READ-ONLY SENSING REUSE PRESERVES PAYLOAD DECISIONS.

CFO (kHz)	Bit errors	BER (%)	Reuse decisions
0	12,700	0.2488	Identical
10	12,891	0.2525	Identical
100	34,540	0.6766	Identical

Projecting both PM columns against all gas/background nuisance terms gives correlation 0.99999962. Scenarios (49, 41) and approximately $(2.45, 98.13) \mu\text{g m}^{-3}$ have optimized-nuisance separation only 4.39×10^{-7} noise standard deviations. The conditional optimal equal-prior binary error is essentially 50%.

Even fixing the other PM mode leaves standard errors of about 29,106 and 35,722 $\mu\text{g m}^{-3}$. Weak absolute sensitivity therefore remains after removing fine/coarse ambiguity. Hypothetical independent fine and total mass observations with 5 $\mu\text{g m}^{-3}$ standard error reduce fine/coarse errors to approximately 5 and 7.07 $\mu\text{g m}^{-3}$. That information comes from the auxiliary observations, not THz retrieval.

V. COMMUNICATION AND ACQUISITION CONTROLS

Communication-only and pilot-reuse sensing keep channel, power and required pilots identical. Modeled net information rates are 783.5117 Mbit/s for the broad reference and 763.4401 Mbit/s for E-band, with zero incremental reuse loss. Extra pilots, retuning or hardware are not assumed free.

A waveform control generates one 10,000-symbol OFDM frame, adds a cyclic prefix and complex time-domain noise, estimates the channel from 30 pilots and decodes uncoded QPSK. Each case scores 5,104,640 payload bits. Sensing reads a copy of the pilot CSI, leaving decoded bits unchanged. The uncoded payload rate is 480.44 Mbit/s, distinct from Shannon information rate. Table III records residual CFO after ideal per-symbol common-phase tracking; no FEC or measured RF response is implied.

Seventeen snapshots of an ideal circular overhead pass recompute refracted background absorption, sky emission, range and instantaneous power allocation. Above 45° , the pass lasts 139.73 s and averages 912.99 Mbit/s in the model. Sampled carrier Doppler reaches 1.211 MHz. This reference excludes Earth rotation, weather availability, FEC and a demonstrated tracking loop.

Separate impairment calculations allow about 29.01 kHz residual CFO or 0.0526 rad RMS phase jitter for a 1% information-rate loss. They are not a joint error budget. WR12 laboratory specifications quote typical 0.1 dB magnitude and 1.5° phase stability after warm-up under stated conditions [10]; these are not a stochastic covariance or receiver calibration recordings.

The E-band design is retained as a concrete communication reference and rejected as the current joint-sensing solution. A laboratory 260–400 GHz extender could examine the aerosol overlap, but a swept VNA scan is not simultaneous OFDM

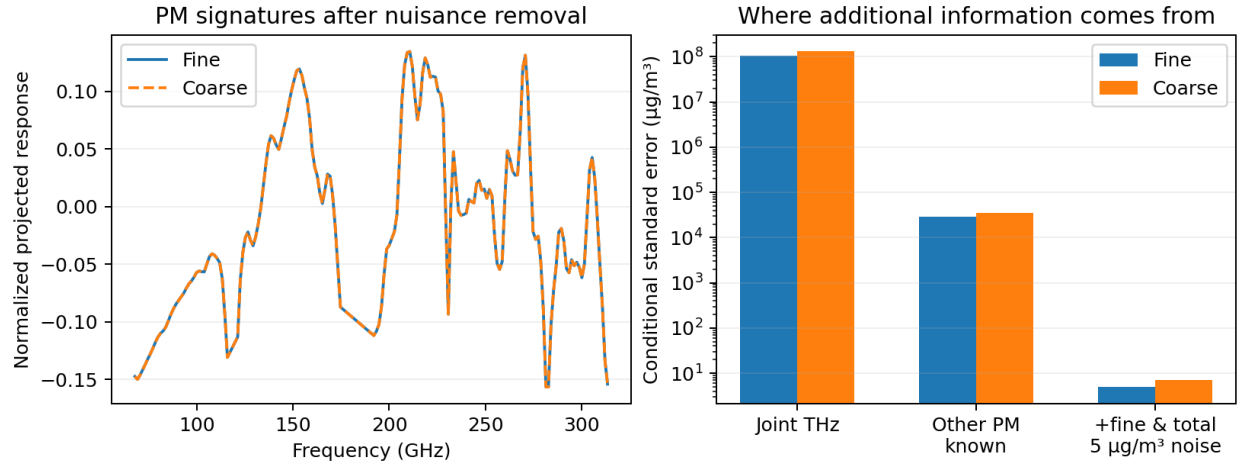


Fig. 1. Projected PM responses and conditional uncertainty. The final bars require hypothetical calibrated mass observations; they are requirements analysis, not acquired data.

acquisition. A useful broad sensing system still needs feasible bands and measured tracking/calibration performance.

VI. INDEPENDENT PUBLIC MEASUREMENT CHECKS

A. Water-background comparison

Taleb et al. report atmospheric-pressure chamber measurements [14]. Their Figure 8 gives 380 GHz humidity/attenuation slopes of 0.033 ± 0.009 for VNA and 0.027 ± 0.009 for THz-TDS in $(\text{dB/m})/(\text{g/m}^3)$. We compare our implementation on a declared unsaturated humidity envelope at six temperatures and each instrument’s reported pressure, without reconstructing their exact regression samples.

The VNA-pressure mean model slope is 0.0328; all six states fall within the reported VNA range. Some TDS-pressure states fall outside its reported range, and that discrepancy is retained. Raw measurements are author-request only. This is an aggregate independent water check, not VOC/PM or satellite validation; no confidence-level interpretation is imposed on the reported plus/minus values.

B. Measured calcite aerosol controls

Eliet’s public dataset provides a one-metre dry-nitrogen calcite cell with corrected THz-TDS means and before/after blanks [15]. Eight sample means and four blanks are reprocessed. Forty-picosecond traces yield approximately 25 GHz native resolution. The source declares useful signal above 300 GHz; only four native bins near 300, 325, 350 and 375 GHz remain within our frequency window. Zero padding is not used to manufacture resolution.

We calculate $-20 \log_{10} |E_s/E_r|$ using geometric-mean blank amplitudes and log-amplitude interpolation between slightly different native grids. Before/after blank changes reach 0.090 dB, while repeat-blank differences stay below 0.01 dB. Across both blank choices, 42 of 64 apparent attenuation values are negative. They are retained as reference-sensitivity evidence, not clipped into particle absorption.

The inspected metadata contain no paired mass concentration or size distribution. Therefore these files cannot establish mass extinction or PM calibration. Calcite in dry nitrogen also differs from mixed ambient aerosol. The result is measured transmission/drift evidence within a limited overlap, not validated $\text{PM}_{2.5}/\text{PM}_{10}$ recovery.

C. VOC evidence screened

An in-band acetonitrile experiment around 165 GHz demonstrates laboratory rotational spectroscopy, but uses low-pressure cells, fitted pressure interpretation and long sequential scans [16]. It cannot validate ambient satellite sensitivity. A located open methanol dataset operates around 3.4 THz and is excluded [17]. No matched atmospheric VOC/PM radio validation dataset was acquired in this bounded search.

VII. ENVIRONMENTAL REQUIREMENTS AND COMPLETION BOUNDARY

WHO concentration guidelines have specific measurement domains and averaging periods [18], [19]; they are not instrument-accuracy specifications. The assessment separates domain, temporal coverage, paired reference data, calibration and conditional sensitivity.

Ambient $\text{PM}_{2.5}/\text{PM}_{10}$ daily values are $15/45 \mu\text{g m}^{-3}$, with annual values $5/15 \mu\text{g m}^{-3}$. Daily assessment also requires its annual percentile context. The 10 s modeled PM limits exceed daily concentration scales by about 3.24×10^7 and 2.45×10^6 . Even an unrealistically continuous independent-noise 24-hour extrapolation gives formal limits near 5.22×10^6 and $1.19 \times 10^6 \mu\text{g m}^{-3}$. These extreme values diagnose missing information, not valid extrapolated sensing ranges.

Formaldehyde’s $100 \mu\text{g m}^{-3}$ reference is a 30-minute indoor guideline. The outdoor slant observation has the wrong domain despite a smaller modeled LOD. An ideal pass above 45° covers only 7.76% of 30 minutes and 0.162% of a day. A stationary 1,800 s control is not continuous coverage or a constellation schedule. The selected WHO sources do

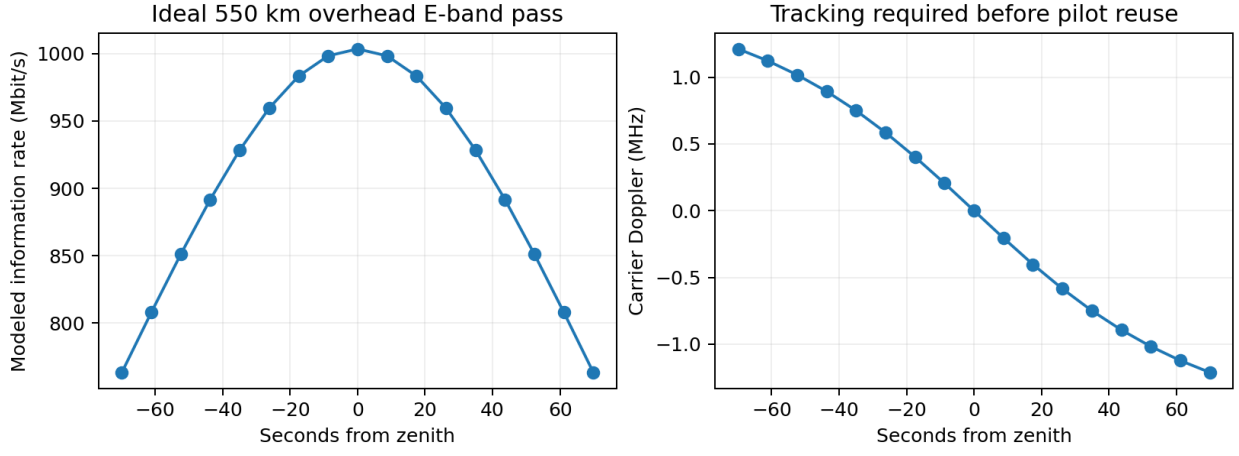


Fig. 2. Ideal moving-pass reference. Same-channel sensing reuse preserves modeled resources; this does not demonstrate tracking, atmospheric availability or useful pollutant retrieval.

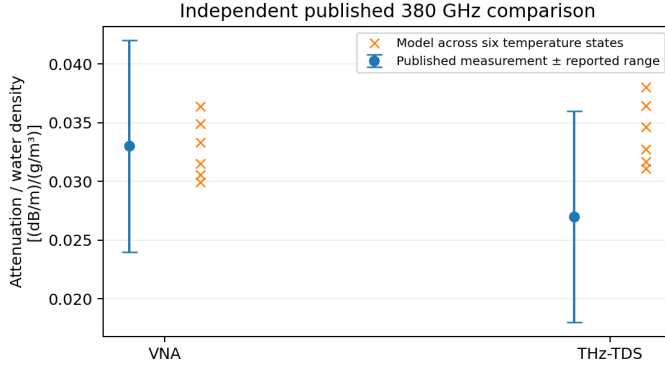


Fig. 3. Published water response and calculated state envelope at 380 GHz. Error bars reproduce reported ranges; crosses are our modeled states.

TABLE IV
FIVE REQUESTED CLOSURE AREAS.

Area	Outcome and remaining boundary
Environmental criteria	Domain/period/detection assessment done; compliance absent.
Practical radio	OFDM, pass and impairment controls done; E-band sensing rejected.
PM limitation	Information diagnosis and measured aerosol re-analysis done; mass calibration absent.
Physical validation	Aggregate water and measured aerosol checks done; paired VOC/PM transmission absent.
Deliverables	Updated manuscript, evidence, scripts and measurement protocol.

not provide ambient methanol/acetonitrile limits; occupational limits are not substituted.

This completes an evidence-based negative/unverified requirements assessment, not compliance verification. Likewise, numerical implementations of atmosphere, spectra, slant integration, PM scattering, CSI, estimators, bounds and declared sensitivity studies do not establish field validation.

VIII. RECOVERING INFORMATION FROM THE COMMUNICATION PAYLOAD

A. A different observable at fixed radio resources

The preceding limits use only 30 of every 10,000 transmitted symbols. In the 10 s broad reference, even perfect knowledge of every other parameter gives local Gaussian recall ceilings of 0.270%, 0.213% and 2.948% at $1 \mu\text{g m}^{-3}$ for the three VOCs. A learned mapping of those same pilot samples cannot supply missing independent information. Instead, we process the already transmitted constant modulus payload.

This direction follows two distinct lessons from the literature. MAPL-EMIT combines physical plume injection, real radiance backgrounds and spatial context, with separate validation selection and physical comparisons [20]. Its methane image recall cannot be transferred to a single radio path. Its use of available information and explicit validation motivates our observable expansion. Payload reuse is already studied in ISAC [21], and second/fourth moment signal and noise separation is established M2M4 estimation [22]. We claim an application and evaluation for atmospheric absorption, not a new moment estimator.

For a single frequency, write $y = \sqrt{S}x + w$, where $|x| = 1$ and $w \sim \mathcal{CN}(0, V)$. Neither the unknown payload bits nor their phase enters the power moments:

$$M_2 = S + V, \quad M_4 = S^2 + 4SV + 2V^2, \quad (8)$$

$$S = \sqrt{2M_2^2 - M_4}, \quad V = M_2 - S. \quad (9)$$

The finite sample implementation replaces $\widehat{M}_2^2$ by $(N\widehat{M}_2^2 - \widehat{M}_4)/(N - 1)$. This removes bias in the estimate of S^2 , not in its square root or logarithm. Invalid inversions remain failures. Absorption is $\widehat{A} = -10 \log_{10}(\widehat{S}/S_{\text{ref}})$.

With $s = S/V$ and $k = 10/\ln 10$, a moment covariance calculation gives

$$\text{var}(\widehat{A}) \simeq \frac{k^2}{N} \left(\frac{2}{s} + \frac{1}{s^2} + \frac{4}{s^3} + \frac{1}{s^4} \right). \quad (10)$$

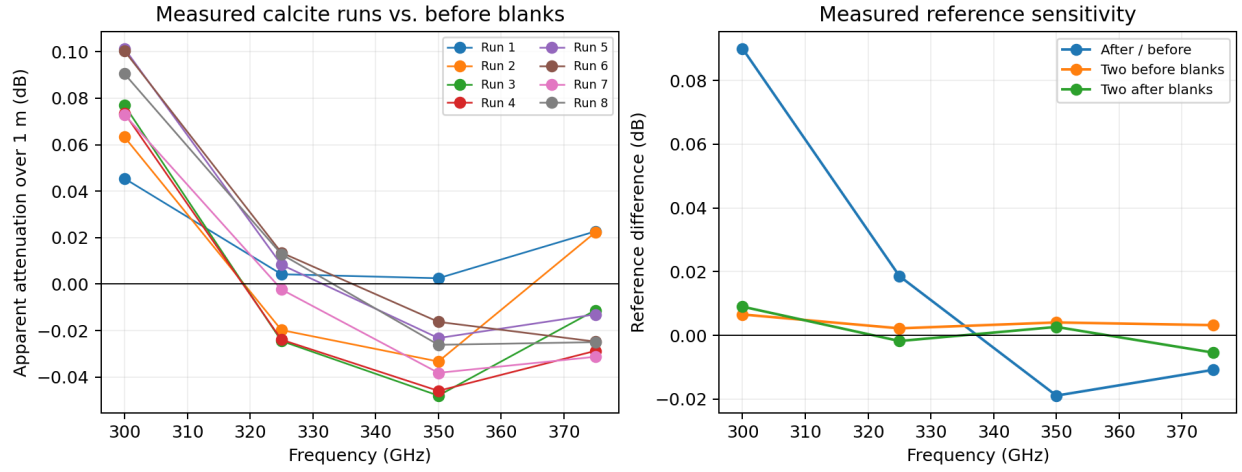


Fig. 4. Measured calcite response and blank sensitivity in four native in-band bins. Lines are visual connections. No mass-concentration labels are available.

The simpler energy method with exactly known noise omits the final two terms. At 5 dB SNR, M2M4's variance is about 1.374 times the coherent known-symbol variance at equal sample count. However, a 10 s window supplies 9,970,000 payload symbols versus 30,000 pilots. The resulting VOC standard errors improve by about a factor of 18. Pilot samples are excluded from the new observable, preventing double counting.

All comparisons retain the previous frequencies, power, joint target design and nuisance columns. They retain signed estimates and the same six-target, 1% family false alarm budget. The broad reference has 1 microsecond symbols and no realizable RF band plan; the separate E-band waveform includes its original cyclic prefix. QPSK reuse preserves the same modem waveform but does not demonstrate attainment of Gaussian-input Shannon capacity.

B. Recall and finite reference results

Each of 12 configurations uses 20,000 positive and 20,000 absent controls. The primary concentration remains $1 \mu\text{g m}^{-3}$; no operating threshold is selected from these outcomes. Table V reports all three VOCs at both durations with zero persistent residual and a calibrated stationary reference magnitude. Acetonitrile's 10 s M2M4 recall has Wilson interval 99.914–99.976%, precision 99.815% at artificial 50% prevalence, and false positive frequency 0.185%. The six-target family false alarm frequency is 0.920%. At hypothetical 1% prevalence, nominal acetonitrile precision is instead about 85.83%.

The 10 s conditional 95%-power VOC limits fall to 3.150, 7.288 and $0.723 \mu\text{g m}^{-3}$. PM remains practically unobservable. An additional experiment allocates 10 s to an independent reference and 10 s to the sample, charging both transmissions equally to both methods. Acetonitrile enhancement recall improves from 0.380% to 93.800%, with total radiated energy 3.991 J. This removes the exactly known reference assumption, while retaining stable atmospheric background. Absolute abundance still requires independently known baseline concentration.

TABLE V
RECALL AT $1 \mu\text{g m}^{-3}$ WITH THE SAME TRANSMISSIONS. ZERO PERSISTENT RESIDUAL, STATIONARY MULTIBAND MODEL.

Time	Target	Pilots (%)	M2M4 (%)
10 s	Formaldehyde	0.265	6.995
10 s	Methanol	0.205	1.085
10 s	Acetonitrile	0.460	99.955
100 s	Formaldehyde	0.415	95.275
100 s	Methanol	0.260	17.220
100 s	Acetonitrile	3.330	100.000

C. Validation and failure boundaries

Pilot means and known-noise energy controls use exact complex Gaussian and noncentral chi-square draws. M2M4 controls use the joint asymptotic distribution of sample moments followed by nonlinear inversion. An independent raw-symbol check uses 240 million QPSK samples over eight settings, including changing phase and a 10% noise change. At 10,000 symbols per trial, measured/predicted variance ratios span 0.970–1.070, with no invalid inversion. Main trials use at least 9.97 million symbols; they are not claimed to simulate every symbol explicitly.

A separate 256-tone FFT/CP replay leaves all 5,104,640 payload bit decisions and the received arrays unchanged after sensing. The uncoded modem retains 12,483 bit errors. Receiver processing is additional even though transmission resources are unchanged. A 1% actual noise-power increase makes the known-noise energy detector's formaldehyde false positive frequency 22.5%; M2M4's tested VOC rates remain below 0.230% across both signs of the noise perturbation.

Persistent calibration is not removed by the extra samples. At the existing 0.001 dB correlated residual, acetonitrile recall falls to 33.310% at 10 s and 48.700% at 100 s. Under arbitrary bounded spectral bias, a robust threshold with radius 0.00001 dB retains 99.84% conditional recall, while radius 0.0001 dB leaves only 15.03%. Those uncertainty models have different meanings.

Leaving each of four public weather profiles out of a nuisance basis produces acetonitrile linearized biases between -0.393 and $+2.007 \mu\text{g m}^{-3}$. These fixed-covariance diagnostics do not support universal recall under changing weather. Applying the PSK moment formula to 16-QAM produces 0.837 dB false attenuation even with infinite data. Appropriate modulation statistics, Gaussian receiver noise and stable or independently normalized gain are necessary. Moments are evaluated on individual subcarriers after the FFT, not on the nonconstant OFDM time waveform.

The existing contiguous E-band remains unsuitable: its all-other-parameters-known local acetonitrile recall ceiling is only 1.256% with 10 s payload reuse. A stationary 100 s reference is not a moving satellite demonstration. Numerical replay checks 436 hashes, estimator identities, confusion counts, variances and resource facts. The evidence in `results/payload_recall` establishes conditional computational improvement, not measured atmospheric recall.

IX. PAYLOAD INFORMATION, SENSITIVITY AND LIMITS

Critical calibration assumption. Favorable results assume zero residual calibration error after correction; this accuracy has not been established experimentally. Receiver noise and finite reference uncertainty remain present. At 45 degrees in the standard atmosphere, with 10 s reference plus 10 s sample, predicted acetonitrile recall at $1 \mu\text{g m}^{-3}$ is 93.84%, 92.55% and 25.17% for differential residual standard deviations of 0, 0.0001 and 0.001 dB, respectively. These are correlated random error scenarios, not measured calibration statistics or worst case drift guarantees. A 0.001 dB power deviation is approximately 0.023%. Persistent systematic drift cannot be removed by longer averaging alone.

A. Unknown data, noise and finite reference

For $y = \sqrt{S}x + w$, equiprobable unit QPSK x and $w \sim \mathcal{CN}(0, V)$, the parameters are $u = \log S$ and $v = \log V$. Each real coordinate has a symmetric BPSK mixture density, connecting the analysis to non data aided estimation [23]. With $a = \sqrt{S}/2$ and $t = 2ar/V$, its scores are

$$g_u = -a^2/V + (ar/V) \tanh t, \quad (11)$$

$$g_v = -1/2 + (r^2 + a^2)/V - t \tanh t. \quad (12)$$

The full QPSK information is $J = 2\mathbb{E}[gg^\top]$. An unknown constant phase is orthogonal to these parameters by symmetry; arbitrary untracked phase instead requires the magnitude observation. For $q = |y|^2$, the Rice likelihood [24] is

$$p(q) = V^{-1} e^{-(q+S)/V} I_0(2\sqrt{qS}/V). \quad (13)$$

Both likelihoods profile unknown receiver noise through $j = J_{uu} - J_{uv}^2/J_{vv}$. With $k = 10/\ln 10$, the local attenuation variance bound is $k^2/(Nj)$. Independent finite sample and reference windows give

$$\text{var}(\hat{A}) \geq k^2 [(N_s j_s)^{-1} + (N_r j_r)^{-1}]. \quad (14)$$

This follows by eliminating their shared unknown baseline power and separate noise powers. A direct joint Fisher inverse

TABLE VI
CHARGED 10 S REFERENCE PLUS 10 S SAMPLE, STANDARD ATMOSPHERE
AT 45 DEGREES, ZERO RESIDUAL. EFFICIENCY IS JOINT FULL-QPSK
CRLB VARIANCE DIVIDED BY M2M4 VARIANCE.

Gas	Limit ($\mu\text{g m}^{-3}$)	Limit (ppm)	Efficiency
Formaldehyde	4.456	0.003509	95.9%
Methanol	10.307	0.007606	97.6%
Acetonitrile	1.023	0.000589	96.5%

verifies it. Equal SNRs favor equal time allocation at fixed total budget.

Per tone information is propagated through the joint concentration and spectral nuisance design using whitened projection and normalized SVD. The information order is known symbols, full QPSK, magnitude, then M2M4, in increasing variance. Numerical quadrature refinement and independent density derivatives verify the scores. These are local regular estimation bounds, not universal detection bounds or biased-estimator MSE guarantees.

M2M4 already retains about 96–98% of the full constellation information in this reference. Estimator replacement therefore offers limited improvement under these assumptions. The limits in Table VI account for concentration dependent sample noise before validation. They are surface equivalent enhancements under the assumed vertical profile. Ppm conversion uses local surface temperature, pressure and natural mixture molar masses.

B. Elevation, weather and changing geometry

The physical sweep recomputes refracted paths, Voigt spectra, background absorption, sky emission, receiver SNR and communication water filling at six elevations from 5 to 90 degrees, for the standard atmosphere and four selected IGRA soundings. All 21 recovered HITRAN isotope files match their previous raw hashes. Durations of 2, 20 and 100 s include equal reference/sample windows. Additional scenarios double receiver noise or introduce 0.0001 and 0.001 dB correlated differential calibration residuals. Frequencies and decision thresholds are fixed.

At nominal receiver noise, 27 of 30 cases retain at least 20 tones above the fixed 5 dB gate. Failed operating points remain explicit. Each eligible 20 s case has 10,000 null responses and 10,000 responses for each VOC at its predicted 95%-power limit. Across 81 controls, observed power is 94.54–95.39%, using nonlinear moment inversion with the previously tested joint moment approximation. Exact binomial intervals and all response arrays are retained. There is no threshold tuning on these trials.

Acetonitrile limits across eligible matched weather cases span 0.287–400.598 $\mu\text{g m}^{-3}$. At standard atmosphere and 20 s, its predicted recall at $1 \mu\text{g m}^{-3}$ changes from 48.17% at 30 degrees to 93.84% at 45 degrees and 99.92% at zenith. The weather supplied to retrieval is correct in this sweep; previously observed weather mismatch biases remain unresolved.

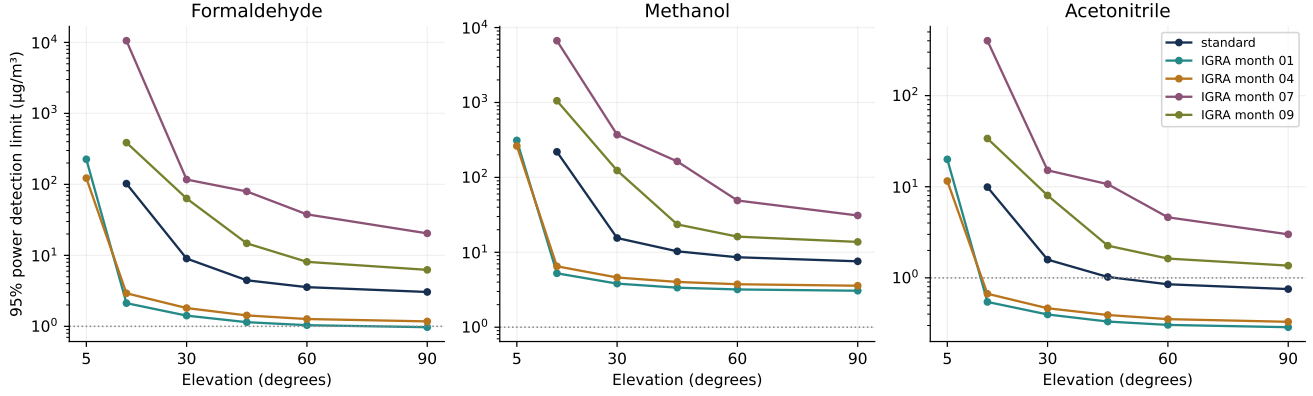


Fig. 5. Matched weather operating region with a charged finite reference. Gaps fail the fixed SNR and usable tone gate. The dotted line is $1 \mu\text{g m}^{-3}$. These are modeled enhancement limits, with zero differential calibration residual and assumed pollutant vertical profiles.

Time dependent information is also integrated along an ideal overhead circular 550 km orbit, with shared concentrations and independent block nuisance coefficients. Matched reference and sample passes both consume transmission resources. At 20 s total, acetonitrile local limits are $1.027 \mu\text{g m}^{-3}$ around 45 degrees and $0.753 \mu\text{g m}^{-3}$ around zenith. Doubling the time quadrature changes standard deviations by less than 0.05%. This assumes accurate instantaneous gain normalization and matched atmosphere; orbital waiting time is excluded. It does not implement synchronization, Doppler tracking or a moving waveform receiver.

C. Standards and unresolved requirements

The WHO 2021 ambient guidelines do not provide methanol or acetonitrile thresholds [18]. Formaldehyde's $100 \mu\text{g m}^{-3}$, 30 minute indoor guideline [19] is not a satellite slant-column compliance threshold. PM_{2.5} and PM₁₀ daily guideline values of 15 and $45 \mu\text{g m}^{-3}$ also differ in averaging time and measurement domain, while modeled PM information remains inadequate. Improved detection limits therefore do not establish environmental compliance.

The added evidence addresses the payload extensions of proposal Tasks 4.1–4.3 under declared assumptions. The principal next requirements are an implementable multiband resource configuration and a calibrated moving receiver, Tasks 2.1–2.2. Joint useful PM retrieval and independent physical concentration validation remain open.

X. SEQUENTIAL RECEIVER AND CALIBRATION FEASIBILITY

A concrete extension replaces simultaneous separated probes with one active RF chain retuned over 16 blocks from 228 to 316 GHz, each with 16 tones at 1 MHz spacing. A one-sample CP, 30 pilots per 10,000 symbols, both reference and sample, 1 ms settling per hop and whole-frame rounding are charged. The 20 s reference/sample budget contains 19.72 s of transmissions. The WR3.4 converter family supports this

frequency conversion range [11], but the assumed 23 dBm radiated power and 6 dB receiver noise figure remain unverified requirements. Typical intrinsic conversion loss and the quoted small-signal input imply about -23 dBm unamplified output before filtering/backoff, leaving at least 46 dB of additional linear gain. This is not demonstrated flight hardware.

Exact Voigt/Mie spectra, refracted paths and emission are recomputed at the selected frequencies. A screened contiguous candidate is rejected after exact joint rank evaluation. With a 0.001 dB correlated differential residual, common gain offset/slope, joint VOC/PM inference and 20 s total, the hopping candidate's response limits are 24.699, 40.176, 4.447 $\mu\text{g m}^{-3}$ for formaldehyde, methanol and acetonitrile. Independent response draws give approximately 95% power at each limit and 0.60% family false alarms. Unknown independent gains at each hop increase acetonitrile's local limit to 548.8 $\mu\text{g m}^{-3}$. Thus retuning requires relative gain calibration, not just a frequency plan.

At $1 \mu\text{g m}^{-3}$ acetonitrile, 100 s total gives 3.14% simulated recall with 0.001 dB residual and 99.99% with 0.0001 dB residual (10,000 positives per case). At 20 s and 0.0001 dB, recall is 90.20% (95% interval 89.60–90.78%). The 100 s result is a favorable *assumed nonzero calibration* control, not measured stability. More averaging does not overcome the persistent residual. The local requirement for 95% power at 100 s is approximately 0.000206 dB under the specified correlation model. Arbitrary systematic drift requires a separate protected-threshold budget.

Raw moving controls now include IFFT/CP generation, integer timing acquisition, CP frequency correction, existing-pilot phase tracking, Hamming (7,4) coding and CRC16. The tested initial Doppler prediction error is 100 kHz; a 700 kHz control fails acquisition and is retained. Three bursts per hop sample changing geometry; only generated symbols enter the sensing variance. Across two reference/sample controls, 66,240 packets decode correctly, and passive sensing preserves decoded output. These two trajectories do not estimate recall.

Hopping/frame scheduling nevertheless costs 1.382% relative to a fixed-band modem with the same code and bandwidth. Processing buffers a 10.625 ms frame. Narrowband Doppler per hop and matched reference weather remain assumed; wideband time dilation, fractional timing, oscillator noise and real-time hardware latency remain unvalidated. PM, absolute baseline truth and field compliance are unresolved.

A. Additional VOCs and PM size fractions

The requested 20 s, 0.0001 dB case was extended using newly acquired HITRAN CH_3Cl and HCOOH lines, with two available isotopologues each. Four other organic candidates remain acquisition gaps. Jointly fitting five VOCs and two PM masses, with the same background/gain nuisance terms and a 1% family budget over eight outputs, reduces simulated $1 \mu\text{g m}^{-3}$ acetonitrile recall to 56.63% in 10,000 joint-mixture responses. Family false alarms are 0.68%. Conditional 95% response limits are $12.114 \mu\text{g m}^{-3}$ for CH_3Cl and $200.879 \mu\text{g m}^{-3}$ for HCOOH , validated at 95.11% and 95.16% simulated recall. Additional species therefore expose ambiguity in the frozen receiver schedule.

PM_{2.5}/PM₁₀ recall at $49/90 \mu\text{g m}^{-3}$ remains only 0.08%, at the false-alarm scale. Even with all other quantities known, a one-parameter fine-PM limit is approximately $905 \mu\text{g m}^{-3}$. Subdividing the fine mode into aerodynamic cuts below and above $1 \mu\text{m}$ produces a numerically unstable three-bin inverse problem: its scaled identity error is 0.508 and the result is rejected. Composition remains assumed; missing particle mass/size calibration cannot be replaced by additional simulated material labels. These results use the same assumed 0.0001 dB residual, matched reference weather and unverified RF requirements. They are not measured field performance.

XI. FIVE-GAS BAND SELECTION AND TIME/CALIBRATION LIMITS

The frozen uniform schedule was revised using a five-gas joint design criterion, retaining both PM masses and all gain/background/interferent nuisance terms. A standard-atmosphere coordinate-exchange screen chose sixteen blocks; exact physical spectra were then recomputed. Neither January weather nor the new response samples selected the bands. Resources remain one sequential RF chain, sixteen 16 MHz blocks, identical pilots/CP, 23 dBm active power, 6 dB noise figure, matched reference, charged settling and equal reference/sample time. A global optimum is not claimed.

At 20 s total and an *assumed*, unmeasured 0.0001 dB residual, selected-design acetonitrile recall at $1 \mu\text{g m}^{-3}$ is 97.67% (exact 95% interval 97.36–97.96%), versus 57.40% for the fresh uniform-design control. The family false-alarm frequency is 0.79%. Every result jointly fits five gases and fine/coarse PM. The nominal 95% response concentrations change with both time and calibration (Table VII); they solve the positive-signal variance equation before Monte Carlo evaluation. PM has no valid 95% response solution within the evaluated ≤ 1 dB absorption domain in any listed condition.

TABLE VII
NOMINAL 95% RESPONSE CONCENTRATION ($\mu\text{g m}^{-3}$), SELECTED FIVE-GAS DESIGN, STANDARD ATMOSPHERE AT 45° , MATCHED REFERENCE. TOTAL TIME INCLUDES REFERENCE AND SAMPLE; σ IS ASSUMED RESIDUAL STANDARD DEVIATION IN dB. PM LIMITS REMAIN UNAVAILABLE IN THE EVALUATED DOMAIN.

Time (s)	σ	H_2CO	CH_3OH	CH_3CN	CH_3Cl	HCOOH
2	0	25.78	27.69	2.92	12.21	102.78
2	0.0001	25.86	27.91	2.94	12.31	102.89
2	0.001	33.02	44.26	4.55	19.69	113.33
20	0	7.57	8.13	0.86	3.58	30.17
20	0.0001	7.85	8.84	0.93	3.92	30.55
20	0.001	21.91	35.38	3.56	15.57	56.27
100	0	3.36	3.61	0.38	1.59	13.40
100	0.0001	3.95	5.00	0.52	2.23	14.23
100	0.001	20.83	34.60	3.47	15.19	49.31

The retained percentage tables report 1, 5 and $10 \mu\text{g m}^{-3}$ VOC recall, low-concentration PM recall, misses, false alarms, exact binomial intervals, relative bias/RMSE and precision/F1 under an explicitly artificial 50% prevalence. A nominal 95% detection point is not a 5% concentration-error point: relative RMSE is approximately 21% there. Zero observed misses do not imply zero population failure probability. Persistent residual covariance is drawn once per acquisition and does not diminish once per payload symbol.

Formaldehyde, methanol, acetonitrile, chloromethane and formic acid are distinct molecular targets; $\text{CO}/\text{O}_3/\text{SO}_2/\text{NO}_2$ remain interferents. PM_{2.5} and coarse PM are disjoint modeled masses, while PM₁₀ is their sum with cross covariance. The proposed PM₁ subdivision remains numerically rejected. Under the assumed material model, smooth Rayleigh absorption is largely removed by gain-slope fitting and leading scattering shapes are nearly shared among size modes. Full Mie calculations do not restore practically useful mass information in these cases.

Fresh moving raw-frame spot checks at each selected frequency preserve decoded packets, but the schedule loses 14.64% of the Gaussian-input rate benchmark relative to the best tested fixed block at equal bandwidth and power. This architecture comparison is distinct from passive observation preserving an already scheduled modem and from coded QPSK throughput. Unknown weather, relative hop gains, realistic timing/oscillator effects, material calibration and independent concentration truth remain open. The results are conditional computational improvements, not validated atmospheric sensor performance.

The common global comparison includes all five VOCs, PM_{2.5}, coarse PM and PM₁₀ in every condition: two band plans, two atmospheres, four elevations, three durations, three calibration residuals and two noise figures (6 and 17 dB), at fixed 23 dBm power. Of 288 conditions, 270 pass and 18 are explicitly rejected for all outputs. Table VIII summarizes the 2304 retained rows. PM₁₀ includes fine/coarse cross covariance. No valid 95% PM limit exists in this grid. These conditional predictions include positive-response variance; their ranges are operating-condition envelopes, not confidence intervals or measured performance.

TABLE VIII

COMMON GLOBAL COMPARISON. REFERENCE CONCENTRATION c IS IN $\mu\text{g m}^{-3}$; RECALL IS PREDICTED PERCENT ACROSS VALID CONDITIONS. N_{95} COUNTS CONDITIONS WITH A RESPONSE LIMIT INSIDE THE EVALUATED ABSORPTION DOMAIN. EACH OUTPUT HAS 270 COMPUTED AND 18 REJECTED CONDITIONS.

Output	c	Recall range (%)	N_{95}
H ₂ CO	1	0.128–89.043	270
CH ₃ OH	1	0.130–72.886	270
CH ₃ CN	1	0.282–100.000	270
CH ₃ Cl	1	0.139–100.000	270
HCOOH	1	0.125–12.651	269
PM _{2.5}	15	0.125–0.125	0
Coarse PM	45	0.125–0.125	0
PM ₁₀	45	0.125–0.125	0

XII. REPRODUCIBILITY AND NEXT EXPERIMENT

Completion arrays reside in results/task_completion; subsequent public measurement analyses are in results/five_task_closure. Payload evidence is in results/payload_recall and results/payload_bounds; the sequential receiver extension is in results/receiver_design. The five-gas design and time/calibration tables are in results/joint_receiver_revision. Table rows are generated from saved CSV/JSON. Source hashes, direct dependency versions, build instructions, tests and numerical checks accompany the revision. Raw data are acquired by script and checked against repository checksums when supplied. Historical failures and manuscripts are preserved.

Software tests do not validate environmental assumptions. Published water agreement cannot determine the complete background covariance. Calcite-cell drift cannot be transplanted into a satellite receiver’s noise law. Four soundings do not define population uncertainty. New diagnostics on previously studied data are not fresh confirmatory field trials.

The next experiment needs calibrated complex transmission; blanks before/after; independent gas concentrations and particle mass/size/composition; weather; timing; and traceable instrument calibration. Calibration and final evaluation campaigns must be separated before inspecting outcomes. The repository supplies a schema and frozen detection/averaging protocol without invented measurements. Those physical milestones remain open without suitable data and hardware.

XIII. CONCLUSION

The pilot study identifies a lack of information, while passive processing of the existing payload supplies a substantial and reproducible conditional VOC improvement. Acetonitrile becomes detectable at $1 \mu\text{g m}^{-3}$ in the original multiband reference under zero residual calibration, including a charged finite-reference experiment. M2M4 is close to the full QPSK information bound in that reference, but the July profile reduces 45 degree acetonitrile recall below 1% at 20 s total even with matched weather. The sequential receiver adds nonzero calibration and coded moving controls; its favorable 100 s

result assumes a 0.0001 dB residual, while retuning has a nonzero scheduling cost. Selecting bands for all five gases raises 20 s acetonitrile recall to 97.67% at $1 \mu\text{g m}^{-3}$ under an assumed 0.0001 dB residual, while a stronger tested fixed-band communication reference exposes a 14.64% Gaussian-rate tradeoff. Neither residual accuracy nor RF requirements are demonstrated experimentally. Methanol, joint PM, arbitrary weather, persistent calibration and the practical E-band case remain unresolved or unfavorable. The next experiment should compare both observables on the same raw QPSK transmission through a controlled gas path, with independent concentration truth and separate reference recordings. Neither simulated recall nor unchanged modem decisions establishes a calibrated satellite pollution monitor.

REFERENCES

- [1] S. Aliaga et al., “Analysis of integrated differential absorption radar and subterahertz satellite communications beyond 6G,” *IEEE J. Sel. Topics Appl. Earth Obs. Remote Sens.*, vol. 17, pp. 19243–19259, 2024, doi: 10.1109/JSTARS.2024.3480816.
- [2] Z. Yang, W. Gao and C. Han, “A universal attenuation model of terahertz wave in space-air-ground channel medium,” *IEEE Open J. Commun. Soc.*, vol. 5, pp. 2333–2342, 2024, doi: 10.1109/OJ-COMS.2024.3386759.
- [3] ITU-R, *Reference standard atmospheres*, Recommendation P.835-7, 2024.
- [4] NOAA NCEI, *Integrated Global Radiosonde Archive*, version 2, Beijing 2026 station archive, accessed September 2026. <https://www.ncei.noaa.gov/products/weather-balloon/integrated-global-radiosonde-archive>
- [5] I. E. Gordon et al., “The HITRAN2024 molecular spectroscopic database,” *J. Quant. Spectrosc. Radiat. Transf.*, 2026, doi: 10.1016/j.jqsrt.2026.109807.
- [6] R. V. Kochanov et al., “HITRAN Application Programming Interface (HAPI): A comprehensive approach to working with spectroscopic data,” *J. Quant. Spectrosc. Radiat. Transf.*, vol. 177, pp. 15–30, 2016.
- [7] ITU-R, *Attenuation by atmospheric gases and related effects*, Recommendation P.676-13, 2022, Annex 1.
- [8] C. F. Bohren and D. R. Huffman, *Absorption and Scattering of Light by Small Particles*. Wiley, 1983. Numerical implementation: miepython 3.0.2.
- [9] ITU-R, *Allocation of ITU-R preparatory work for WRC-27*, Resolution 178 discussion, Circular Letter CA/251.
- [10] Virginia Diodes, *VNA Frequency Extension Modules: Summary of Performance Specifications*, VDI-956, rev. March 17, 2022, accessed September 2026.
- [11] Virginia Diodes, *Compact Converter Operational Manual*, VDI-737, rev. August 27, 2024, pp. 6–9, accessed September 27, 2026. https://vdiodes.com/wp-content/uploads/2012/01/VDI-737_CC_Product_Manual.pdf
- [12] S. M. Kay, *Fundamentals of Statistical Signal Processing: Estimation Theory*. Prentice Hall, 1993.
- [13] S. Chen, “Beijing Multi-Site Air Quality,” UCI Machine Learning Repository, 2017, doi: 10.24432/C5RK5G.
- [14] F. Taleb et al., “Propagation of THz radiation in air over a broad range of atmospheric temperature and humidity conditions,” *Scientific Reports*, vol. 13, art. 20782, 2023, doi: 10.1038/s41598-023-47586-8.
- [15] S. Eliet, “TeraHertz time domain spectroscopy of calcite particles resuspended (aerosol form),” Recherche Data Gouv, version 1, 2026, doi: 10.57745/DLJEFW. Acquired September 11, 2023; authors’ corrected outputs use Correct@TDS.
- [16] D. Gudž et al., “(Sub-)millimeter spectroscopy instrument for the soil metabolite detection,” *IEEE Sensors J.*, vol. 25, no. 22, pp. 41274–41281, 2025, doi: 10.1109/JSEN.2025.3614491.
- [17] M. D. Horbury et al., *Data associated with Real-Time Terahertz Absorption Spectroscopy of Methanol and Deuterated-Methanol Vapour, using a TeraFET Detector Array*, University of Leeds, doi: 10.5518/1321. Screened and excluded from validation because it is out of band.
- [18] World Health Organization, *WHO Global Air Quality Guidelines*, 2021, ISBN 9789240034228.

- [19] World Health Organization, *WHO Guidelines for Indoor Air Quality: Selected Pollutants*, 2010, formaldehyde chapter, ISBN 9789289002134.
- [20] V. V. Batchu et al., “Global monitoring of methane point sources using deep learning on hyperspectral radiance measurements from EMIT,” *PNAS*, vol. 123, no. 36, e2612145123, 2026, doi: 10.1073/pnas.2612145123.
- [21] C. Xu, X. Yu, F. Liu and S. Jin, “Exploiting both pilots and data payloads for integrated sensing and communications,” arXiv:2506.15998, 2025.
- [22] G. Romano, “On asymptotic efficiency of the M2M4 signal-to-noise estimator for deterministic complex sinusoids,” *Sensors*, vol. 21, no. 15, art. 4950, 2021, doi: 10.3390/s21154950.
- [23] F. Brännström and L. K. Rasmussen, “Non-data-aided parameter estimation in an additive white Gaussian noise channel,” arXiv:cs/0509007, 2005.
- [24] H. Gudbjartsson and S. Patz, “The Rician distribution of noisy MRI data,” *Magn. Reson. Med.*, vol. 34, pp. 910–914, 1995, doi: 10.1002/mrm.1910340618.